Register Number:

Name of the Candidate:

0319

B.E. DEGREE EXAMINATION, 2022

(COMMON TO CS, IT, AND AI & ML)

(FOURTH SEMESTER)

CSBS–401/ITBS-401/AIBS-401: DISCRETE MATHEMATICS

(For those who joined from 2018-2019)

June] [Time : 3 Hours

Maximum : 75 Marks

SECTION – A

Answer ALL questions **(10 × 1 =10)**

		Marks	CO	PO	BTL
1.	Define a contradiction of a logical statement	1	CO1	1,2	K1
2.	Define the conjunction of two logical statements P and Q	1	CO1	2,3	K2
3.	Define universal relation with example	1	CO2	1,2	K1
4.	Define equivalence relation with an example	1	CO2	1,2	K1
5.	Define a complemented Lattice	1	CO3	2,3	K1
6.	State absorption laws of Boolean algebra	1	CO3	1,2	K1
7.	Define a semi group with an example	1	CO4	1,2	K1
8.	what is the identity element of the algebraic system $\{Q^+,*\}$ Where Q^+ is the set of positive rational numbers and * is defined by $a * b = \frac{ab}{2}$ for all $a, b \in Q^+$	1	CO4	3,4	K2
9.	Define complete graph with an example	1	CO5	1,2	K1
10.	Define spanning subgraph.	1	CO5	1,2	K1

SECTION – B

Answer either (a) or (b) in each Question **(5 × 3 =15)**

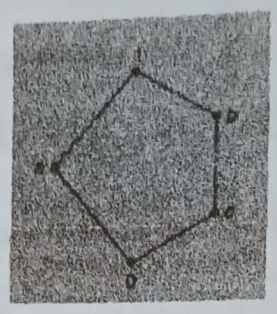
			Marks	CO	PO	BTL
11.	(a)	Show the implication without constructing the $(P \rightarrow Q) \rightarrow Q \Rightarrow P \vee Q$	3	CO1	1,2	K2
		(OR)				
	(b)	Obtain the PDNF and PCNF of $(\neg P \vee \neg Q) \rightarrow (P \leftrightarrow \neg Q)$	3	CO1	2,3	K2
12.	(a)	If R and S are equivalence relations on A, show that $R \cap S$ is also an equivalence relation	3	CO2	2,3	K2
		(OR)				

(b) If $A = \{1,2,3\}$ and $R = \{(1,1),(1,2),(2,1),(2,3),(3,1),(3,2)\}$ $S = \{(1,1),(1,2),(2,1),(2,2),(3,2),(3,3)\}$, be a relations defined on A. Find $R \circ S$, $R \oplus S$ and R^2

3	CO2	2,3	K2

13. (a) Verify whether the lattice given by the Hasse diagram is distributive

3	CO3	1,2	K2



(OR)

(b) Simplify the Boolean expression $xy' + z + (x' + y)z'$

3	CO3	2,3	K2

14. (a) Prove that $\{Z_6, \times_6\}$ is a monoid

3	CO4	1,2	K2

(OR)

(b) Show that $\{(1,-1,i,-i),\times\}$ is a cyclic group. What are its generators?

3	CO4	3,4	K2

15. (a) Define Eulerian and Hamiltonian graphs with example

3	CO5	1,2	K2

(OR)

(b) Draw the graph whose adjacency matrix is $\begin{pmatrix} 1 & 2 & 0 & 1 \\ 2 & 0 & 3 & 0 \\ 0 & 3 & 1 & 1 \\ 1 & 0 & 1 & 0 \end{pmatrix}$

3	CO5	2,3	K2

SECTION – C

Answer either (a) or (b) in each Question **(4 × 10 =40)**

16. (a) Find whether the conclusion C: R follows from the premises $H_1: P \vee Q$; $H_2: \neg R \rightarrow \neg P$; $H_3: \neg(Q \wedge \neg R)$ using truth table technique

10	CO1	2,3	K3

(OR)

(b) Show that $t \wedge s$ can be derived from the premises $(p \rightarrow q)$, $q \rightarrow \neg r$, r, $p \vee (t \wedge s)$

10	CO1	3,4	K3

17. (a) Show that for any three sets $A \times (B \cup C) = (A \times B) \cup (A \times C)$

10	CO2	2,3	K3

(OR)

(b) If R is the relation on the set of ordered pairs of positive integers such that $(a,b) \in R$ if and only if $3a + 4b = 7n$ for some integer n Show that R is an equivalence relation

10	CO2	1,3	K3

18. (a) Show that $(a + b')(b + c')(c + a') = (a' + b)(b' + c)(c' + a)$ in a Boolean algebra.

10	CO3	1,2	K3

(OR)

(b) Simplify the Boolean function $f(a,b,c,d) = \sum(0,1,2,3,4,5,6,7,8,9,11)$

10	CO3	2,3	K3

19. (a) Prove that the set {1,3,7,9} is an abelian group under multiplication modulo 10 | 10 | CO4 | 2,3 | K3 |

(OR)

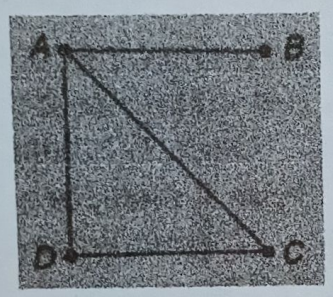
(b) List the code words generated by the encoding function | 10 | CO4 | 2,3 | K3 |

e: $B^3 \rightarrow B^6$ whose parity check matrix is $H = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

SECTION – D

Answer either (a) or (b) (1 × 10 =10)

20. (a) Summarize all the paths of length 4 between B and D for the following graph using adjacency matrix | 10 | CO5 | 2,3 | K4 |



(OR)

(b) Evaluate the shortest distance matrix and the corresponding shortest path matrix for all the pairs of vertices in the weighted graph given below using Warshall's algorithm. | 10 | CO5 | 2,3 | K4 |

